

Acemoglu & Restrepo (2018)

The Race between Man and Machine

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`github.com/carlosgomezp28/ai-06-restrepo`

Question. How do automation and the creation of new tasks affect output, wages, employment, and factor shares?

Key shift relative to the other papers.

- ▶ The unit of analysis is the **aggregate economy**, not an individual worker or AI agent.
- ▶ Technology changes *which tasks* capital and labor perform.
- ▶ The central race is between **displacement** and **reinstatement**.

Static environment. A unit measure of tasks is indexed by

$$i \in [N - 1, N].$$

2 · The firm's task-level problem

Labor productivity in task i is $\gamma(i)$, with $\gamma(i)$ strictly increasing.

If task i is technologically automatable, the firm compares

$$R \quad \text{vs.} \quad \frac{W}{\gamma(i)}.$$

- ▶ R : rental rate of capital.
- ▶ $W/\gamma(i)$: effective unit cost of labor in task i .
- ▶ Higher-index tasks are those in which labor has stronger comparative advantage.

Agent's problem: competitive firms assign each task to the lowest-cost feasible factor.

3 · Technology and the equilibrium cutoff

Automation technology is summarized by $I \in [N - 1, N]$:

$i \leq I \Rightarrow$ capital is technologically feasible,

$i > I \Rightarrow$ labor only.

The cost-indifference threshold $\tilde{I}$ solves

$$\frac{W}{R} = \gamma(\tilde{I}).$$

Therefore equilibrium automation is

$$I^* = \min\{I, \tilde{I}\}.$$

$i \leq I^*$: capital

$i > I^*$: labor

4 · Conditions behind the static results

The main static propositions impose Assumptions 1–3.

1. $\gamma(i)$ is strictly increasing.
2. Either

$$\eta \rightarrow 0 \quad \text{or} \quad \zeta = 1.$$

This delivers homothetic factor demands in the baseline exposition.

3. $K < \bar{K}$, where $\bar{K}$ is defined by

$$R = \frac{W}{\gamma(N)}.$$

Hence in the maintained region

$$R > \frac{W}{\gamma(N)},$$

so newly created tasks are profitable to use immediately.

Under these conditions the static equilibrium exists and is unique.

Consider

$$I^* = I < \tilde{I}.$$

The range of automated tasks is constrained by available technology.
For automation,

$$\frac{d \ln(W/R)}{dI} = -\frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} < 0.$$

Sign implications

A rise in I reduces W/R , the labor share, and employment.

The sign of the **real wage** W is not pinned down by this result alone.

6 · When more automation technology does nothing

Now consider

$$I^* = \tilde{I} < I.$$

Firms are already choosing the cost-minimizing boundary.

More technically automatable tasks do not change the allocation at the margin.

Therefore a marginal increase in I has no effect on equilibrium factor prices.

Technological feasibility $I \neq$ equilibrium automation I^*

Automation: displacement

- ▶ Expands tasks performed by capital.
- ▶ Contracts the task range performed by labor.
- ▶ Pushes relative labor demand downward.

New tasks: reinstatement

- ▶ Raises N .
- ▶ Introduces tasks where labor has comparative advantage.
- ▶ Raises relative labor demand.

The long-run model is a race between these two margins.

Under Assumptions 1–3, in the technology-constrained regime,

$$d \ln W = d \ln Y|_{K,L} + (1 - s_L) \left[\frac{\Lambda_N}{\widehat{\sigma} + \varepsilon_L} dN - \frac{\Lambda_I}{\widehat{\sigma} + \varepsilon_L} dI \right].$$

For a pure automation shock, $dN = 0$ and $dI > 0$:

$$d \ln W = \underbrace{d \ln Y|_{K,L}}_{\text{productivity (+)}} - \underbrace{(1 - s_L) \frac{\Lambda_I}{\widehat{\sigma} + \varepsilon_L} dI}_{\text{displacement (-)}}.$$

9 · The week's trap: does automation necessarily lower wages?

No.

Automation in the technology-constrained regime always lowers W/R , but

$$dW > 0 \iff \text{productivity effect} > \text{displacement effect}.$$

- ▶ **Productivity:** replacing expensive labor with cheaper capital raises output holding K, L fixed.
- ▶ **Displacement:** labor is squeezed into a smaller set of tasks.

So $W \uparrow$ can coexist with

labor share $\downarrow$, employment $\downarrow$.

Still under Assumptions 1–3 and

$$I^* = I < \tilde{I},$$

Proposition 3 implies that there exists a finite threshold $\tilde{K}$ such that

$$K > \tilde{K} \implies \frac{dW}{dI} > 0,$$

while

$$K < \tilde{K} \implies \frac{dW}{dI} < 0.$$

Interpretation. A larger capital stock makes the cost-saving/productivity channel from automation strong enough to dominate displacement.

- ▶ Reconstructed the task-allocation logic: I , $\tilde{I}$, and I^* .
- ▶ Re-derived by hand why

$$\frac{d \ln(W/R)}{dI} < 0$$

does *not* imply

$$\frac{dW}{dI} < 0.$$

- ▶ Checked the productivity-versus-displacement condition and the $\tilde{K}$ threshold.
- ▶ Ran my own AppliedModelingLib formalization as AR18RaceManMachine.
- ▶ Repaired three proof/build issues without changing statements or economic assumptions.
- ▶ Final paper-scoped check: 832 jobs, exit code 0.

Computational contribution: formal verification of selected reduced-form comparative-statics claims, with an auditable build/check trail.

12 · Where I did not believe the AI: hand check I

Perkins &

Automation and Wages

$$i \in \{N, S, M, L\}$$

$$I^* = \min\{I, \bar{I}\}$$

$$i \leq I^* \Rightarrow \text{task performed by capital}$$

$$i > I^* \Rightarrow \text{task performed by labor}$$

Technology - constant return:

$$I^* = I < \bar{I}$$

An increase in I expands the set of tasks performed by capital

$$\frac{d \ln(W/R)}{dI} = -\frac{\Delta_L}{\Delta_L \Delta_K} < 0$$

Therefore,

$$dI > 0 \Rightarrow d \ln(W/R) < 0$$

Displacement effect: automation reallocate tasks away from labor, reduces labor's relative demand

But a fall in W/R does not necessarily imply a fall in W

$$d \ln W = \underbrace{d \ln Y|_{KL}}_{\text{Productivity effect}} + \underbrace{(1-S_L) d \ln C|_{KL}}_{\text{Displacement effect}}$$

Using the automation effect:

$$d \ln W = d \ln Y|_{KL} - (1-S_L) \frac{\Delta_L}{\Delta_L \Delta_K} dI$$

Hence,

$$dW > 0 \Leftrightarrow d \ln Y|_{KL} > (1-S_L) \frac{\Delta_L}{\Delta_L \Delta_K} dI$$

So:

Productivity effect > Displacement effect $\rightarrow W \uparrow$
 Productivity effect < Displacement effect $\rightarrow W \downarrow$

Question I checked: Does $\frac{d \ln(W/R)}{dI} < 0$ imply $\frac{dW}{dI} < 0$? **No: the first is a relative-price result.**

13 · Where I did not believe the AI: hand check II

Proposition 3:	$\exists \bar{R} < \infty$
such that	$R > \bar{R} \rightarrow d \ln W / d \ln I > 0$
and	$R < \bar{R} \rightarrow d \ln W / d \ln I < 0$
Verdict	
Information: thanks to my powers, I'll be in the industry-adjusted future, but if demand decreases it's lower the real wage. The sign of the wage effect depend on whether productivity gains dominate displacement.	
Conclusion	W/R
Conclusion can still involve	labor share $\downarrow$, employment $\downarrow$

Verdict: $d \ln W =$ productivity effect — displacement effect.

The sign of dW is ambiguous even though W/R , the labor share, and employment fall.

Source-facing mathematical core of Proposition 3

$$\text{cost saving} > 0 \implies \text{productivity contribution} > 0.$$

In the reduced-form formalization,

$$\text{productivityContribution} = \text{scale}(\text{expensive} - \text{cheap}).$$

Hence the checked sign claim is

$$\text{scale} > 0, \quad \text{cheap} < \text{expensive}$$

$$\implies \text{productivityContribution} > 0.$$

The wage and rental decompositions are represented separately as exact algebraic identities.

15 · Lean: statement and proof fragment

```
theorem productivityContribution_pos
  {scale expensive cheap : ℝ}
  (hscale : 0 < scale)
  (hcost : cheap < expensive) :
  0 < productivityContribution scale expensive cheap := by
  exact mul_pos hscale (sub_pos.mpr hcost)

theorem proposition3_productivity_factor_price_core :
  proposition3_productivity_factor_price_coreSpec := by
  intro scale expensive cheap productivity laborShare
  relativePriceChange hscale hcost
  exact ⟨
    productivityContribution_pos hscale hcost,
    wageChangeDecomposition_eq
      productivity laborShare relativePriceChange,
    rentalChangeDecomposition_eq
      productivity laborShare relativePriceChange
  ⟩
```

- ▶ **Objects:** real-valued reduced-form quantities for scale, costs, productivity, labor share, and relative-price change.
- ▶ **Nontrivial proof step:** `sub_pos.mpr hcost` proves

$$\text{expensive} - \text{cheap} > 0,$$

and `mul_pos` combines it with positive scale.

- ▶ **Exact identities:** the wage and rental decompositions close by definitional equality in the underlying model file.
- ▶ **Boundary:** Lean does *not* derive these reduced-form quantities from the full continuum-task equilibrium or prove the full $K \gtrless \tilde{K}$ threshold.

Final check

Build completed successfully (832 jobs), `exit_code=0`.

- ▶ A successful build is only meaningful relative to the statement that was encoded.
- ▶ The paper's economic content lives in the bridge from the task economy to the reduced-form comparative statics.
- ▶ Lean is strongest here at exposing exactly which sign conditions are assumed and which conclusions follow algebraically.
- ▶ The main remaining formalization challenge is therefore **economic**: formalize the continuum task allocation and derive Proposition 3 rather than starting from its reduced form.

1. Automation is not just factor-augmenting technical change: it reallocates tasks from labor to capital.
2. In the technology-constrained regime, automation lowers W/R , the labor share, and employment.
3. The real wage is ambiguous because productivity and displacement push in opposite directions.
4. New tasks generate reinstatement and push labor demand in the opposite direction.
5. The long-run question is therefore not simply “man or machine?” but which margin advances faster: automation or new labor-intensive tasks.

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